

# BitStreamer User Manual

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Welcome to BitStreamer! This manual covers everything you need to know to set up, run, and troubleshoot the BitStreamer server and client.

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## 1. Overview

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BitStreamer is a Free, open-source, extremely lightweight, zero-transcode **single file or multi-folder** local network media **streamer** with it's own lightweight **client App** that can be **sideloaded into FireTV**. You can find both the server and the client code in the github sources and the binaries in the release. It uses the native Exoplayer and supports bitstreaming of Audio codecs including Dolby TrueHD, Dolby Atmos and DTS-HD (only DTS core because of FireTV limitations). The server also serves the client APK to be used from Downloader app in FireTV for sideloading. No need to connect to external site for this. You can build and run the whole code from your local machine if you have the right tools.

- **The Server** (run on your PC/Mac/Linux box) serves your media file byte-for-byte over HTTP plus the client apk. It also supports generation of Chapter and seek-bar thumbnails/preview (like in netflix/youtube) if you have ffmpeg and ffprobe executibles in the path or the same folder as server (see [Chapters & Scrubbing Previews](#) below). Also supports external subtitles with same filename as the movie file.
- **The Client** (run on your Fire TV) discovers the server automatically, plays the file with hardware video decoding, and bitstreams the audio untouched over HDMI to your TV/AV receiver. It also has corresponding features to the server like chapter thumbnails, seek-bar thumbnails depending on whether server has ffmpeg/ffprobe in the PATH/dir. Remember to press the **Menu (≡)** in different places in the app to see options

### Note:

- Folder support is fully completed with v1.5.0! You can run the server on a single folder, a drive root, or **multiple directories simultaneously** ( `bitstreamer.exe "C:\Movies" "D:\TV Shows"`  ) to browse, select root directories, play videos, run slideshows of images, view chapters, and resume playback per file directly from the Fire TV client.
  - Also check [Known Issues](#) below to know what the Hardware itself doesn't support.
  - This is untested on Nvidia Shield but might work there since it is built for Android TV - if someone does, let me know how it goes.
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## 2. Server Setup & Usage

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### Running the Server

On your PC, open a command prompt/terminal and run the server by passing either a single media file, a folder directory, or multiple folder directories:

```
# Windows
bitstreamer.exe "C:\Movies\your-movie.mkv"           # Single-file mode
bitstreamer.exe "C:\Movies"                          # Folder mode (single folder)
bitstreamer.exe "D:\"                               # Serve a drive root
bitstreamer.exe "C:\Movies" "D:\TV Shows" "E:\Videos" # Multi-folder mode (multiple roots)

# macOS
./bitstreamer-macos "/Volumes/Movies/your-movie.mkv" # Single-file mode
./bitstreamer-macos "/Volumes/Movies" "/Volumes/TV"  # Multi-folder mode
```

```
# Linux
./bitstreamer-linux "/path/to/your-movie.mkv"          # Single-file mode
./bitstreamer-linux "/path/to/movies" "/path/to/tv"    # Multi-folder mode
```

The server will print:

- Your local network IP addresses
- The streaming URL (for checking in a browser)
- The Client APK download link

### Windows Firewall Setup (One-Time)

If you run the server on Windows, you must allow incoming traffic. Run these commands as **Administrator**:

```
netsh advfirewall firewall add rule name="BitStreamer HTTP" dir=in action=allow protocol=TCP localpor
netsh advfirewall firewall add rule name="BitStreamer Discovery" dir=in action=allow protocol=UDP loc
```

### Advanced Server Options

- `--port 46898` : Set a custom port for the HTTP server.
- `--name "MyServer"` : Set a custom display name announced to clients.
- `--interval 30` : Set the scrubbing-preview interval in seconds (default: 30, automatically reduces to 10 for videos under 10 minutes).
- `-stripdv` : Forces client-side Annex B stripping of Dolby Vision metadata to fallback to HDR10 (only applies to Profile 7 files).
- `-skip-previews` : Skips generating the storyboard seek-bar preview sprites at startup.
- `--no-caching` : Disables persistent disk caching of metadata and thumbnails in `cache/` (chapter discovery still works).
- `--version` : Print the server version and exit.

### HDR & Dolby Vision Playback Behavior

| HDR Format                | Supported? | Behavior / Handling                                                                              |
|---------------------------|------------|--------------------------------------------------------------------------------------------------|
| SDR (BT.709)              | ✔ Yes      | Direct native playback.                                                                          |
| HDR10 (PQ, BT.2020)       | ✔ Yes      | Direct native playback.                                                                          |
| HDR10+ (Dynamic metadata) | ✔ Yes      | Direct native playback with dynamic metadata mapped.                                             |
| HLG (Hybrid Log Gamma)    | ✔ Yes      | Direct native playback.                                                                          |
| Dolby Vision Profile 5    | ✔ Yes      | Direct native playback (IPTPQc2 colorspace).                                                     |
| Dolby Vision Profile 8.1  | ✔ Yes      | Direct native playback (even if file also carries HDR10+ metadata, e.g. <i>Ford v Ferrari</i> ). |

|                                   |       |                                                                                                                                                                                                            |
|-----------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dolby Vision Profile 7 MEL</b> | ✓ Yes | Direct native playback of the base layer which is a standard HDR10/HDR10+ stream.                                                                                                                          |
| <b>Dolby Vision Profile 7 FEL</b> | ✗ No  | <b>Audio plays, video remains black.</b> Fire TV hardware cannot decode dual-layer FEL. Plays using our zero-allocation NAL stripping ( Strip DV and Play / -stripdv ), or Convert to DV8 and Play option. |

### 3. Client Installation & Usage

#### Installing the Android TV Client

##### Method 1: Command Line via ADB (Fastest)

If you have `adb` installed on your PC and ADB Debugging enabled on your Fire TV (*Settings -> My Fire TV -> Developer Options -> ADB Debugging -> ON*): (Tip: Find your Fire TV's IP address under **Settings -> My Fire TV -> About -> Network**).

1. Run `install-client.bat` from the command line or double-click it in Windows Explorer:

```
install-client.bat 192.168.1.50
```

2. Or use standard ADB commands directly:

```
adb connect 192.168.1.50:5555
adb install -r client.apk
```

##### Method 2: Downloader App on Fire TV

1. On your Fire TV, search for and install the **Downloader** app from the Amazon Appstore.
2. Open Downloader and enter the APK URL printed by your server (e.g., `http://192.168.1.20:46898/client.apk` ).
3. Download and install the APK. (Note: You will need to allow Downloader to "Install unknown apps" in Fire TV Settings).

#### Playing Media

1. Launch **BitStreamer** on your Fire TV.
2. The app will search for servers on the local network. Select the found server.
3. In Multi-Folder mode, select your desired root folder from the Root Selection screen.
4. If the server is not found automatically, enter your PC's IP address manually in the input box at the bottom.
5. **Resume:** When you re-open a video you previously watched partway, BitStreamer prompts to **Resume** from where you left off or start over. Positions are stored per file on the server ( `resume.json` ) and cleared automatically once you finish a title ( $\geq 90\%$ ).

#### Remote Control Mapping

- **List Navigation Mode:**
  - **Center / OK:** Enter directory, select root, or start file playback.

- **Back:** Go back up one directory level (retains selection memory).
- **Menu (≡):** Open the file's context menu. Dolby Vision files offer `Play Normally` , `Strip DV` and `Play` , and — in folder mode — `Convert to DV8` and `Play` ; non-Dolby-Vision files just offer `Play Normally` . In folder mode a `Generate Seekbar Previews` checkbox also appears when previews are available.
- **Video Playback Mode:**
  - **Center / Play-Pause Button:** Play or pause the video.
  - **D-pad Left / Right:** Jump backwards/forwards by one preview interval (default 30s, automatically smaller for short videos) and show the scrubbing preview frame.
  - **Rewind / Fast Forward Buttons:** Jump to the previous or next chapter directly (falls back to standard seek if no chapters).
  - **D-pad Up:** Focus and display the seek bar.
  - **D-pad Down:** Open the options menu (Audio, Subtitles, Chapters, Stats).
  - **Previews Button (Film Strip Icon):** Trigger on-demand seek-bar preview generation with real-time progress bar UI.
  - **Menu Button:** Toggle the **Stats for Nerds** overlay — live bandwidth, video/audio codecs and bitrate, resolution, HDR/colour, the active decoder, and the Dolby Vision conversion in effect (e.g. `dv7` → `dv8` , `dv7` → `hdr10` , or `Native DV` ).
  - **Back Button:** Hide controls or exit playback (prevented during preview generation).
- **Image Playback Mode:**
  - Plays each image for **5 seconds** and auto-advances.
  - **Center Button:** Pause or resume the slideshow auto-advance.
  - **Rewind / Fast Forward Buttons:** Go to the previous or next image in the directory immediately.
  - **Back Button:** Exit image playback.

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## 4. Getting Best Audio & Video

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### Audio Bitstreaming (HDMI Passthrough)

To ensure raw compressed audio is handed directly to your soundbar/receiver:

1. On your Fire TV, go to **Settings** → **Display & Sounds** → **Audio** → **Surround Sound**.
2. Select **Best Available** (or **Dolby Digital Plus / Atmos** depending on your OS version).
3. Check your AV Receiver or TV display panel to confirm it shows the correct audio stream (e.g., *Dolby Atmos*, *DTS*, *Dolby Digital*).

[!NOTE] **DTS-HD** streams are played as **DTS Core** (5.1 surround sound extracted on the fly), since Fire TV hardware does not support full DTS-HD passthrough. **Dolby TrueHD** is supported only on newer models like the Fire TV Stick 4K Max (2nd Gen).

### Subtitles

- BitStreamer automatically detects sidecar subtitle files (e.g., `.srt` , `.ass` , `.vtt` ) placed next to the movie file on the server.
- Use the D-pad Down menu to toggle subtitles on/off or change tracks.

### Chapters & Scrubbing Previews (Optional)

To enable visual chapter markers and Netflix-style scrubbing previews:

1. Download `ffmpeg` and `ffprobe` binaries [link](#).
2. Place `ffmpeg.exe` and `ffprobe.exe` (or your OS equivalents) in the same directory as the `bitstreamer` server executable, or put them on your system's `PATH` .

- Chapter thumbnails are generated server-side.
  - Seekbar previews** are on-demand: request preview generation either from the file launch menu ( `Generate Seekbar Previews` checkbox) or in-player via the **Previews** (film strip) control button. A progress bar overlay tracks progress on-screen, automatically pausing playback until generation completes.
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## 5. Known Issues

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### DTS-HD / DTS:X Playback (Limited to DTS Core)

- Issue:** Lossless DTS-HD Master Audio and DTS:X audio tracks are played as standard **DTS Core** (5.1 surround sound) instead of full lossless audio.
- Cause:** This is a hardware limitation of the Fire TV platform. Fire OS does not expose the full DTS-HD lossless bitstream to third-party Android apps via standard `AudioTrack` APIs; it only makes the 5.1-channel DTS Core portion reachable.
- Workaround:** There is no direct app workaround. If you require lossless audio, a device that supports full Dolby TrueHD and DTS-HD passthrough (such as Nvidia Shield TV Pro) is required.

### Dolby Vision Profile 7 FEL (Black Screen, Audio Only)

- Issue:** When attempting to play a Dolby Vision **Profile 7 FEL** (Full Enhancement Layer) dual-layer video track (commonly found on 4K UHD Blu-ray REMUX files), the audio plays perfectly but the **screen remains black**.
- Cause:** Dual-layer Profile 7 files contain a Base Layer (HDR10 compatible) plus a Full Enhancement Layer (FEL) containing actual picture residual details. The Fire TV's hardware Dolby Vision decoder cannot decode the enhancement layer and stalls. (Note: Profile 7 **MEL** or Profile 8 files play natively as Dolby Vision without issues).
- Workaround:**
  - On-the-Fly NAL Stripping:** Use the `-stripdv` server-side flag, or press **Menu** ( $\equiv$ ) on the remote in the list view and choose `Strip DV and Play` to decode the stream on the fly as standard HEVC/HDR10 with zero-allocation high-performance NAL stripping.
  - On-the-Fly DolbyVision 8 Conversion:** Press **Menu** ( $\equiv$ ) on the remote and choose `Convert to DV8 and Play` to remap the Dolby Vision metadata to Profile 8 on the fly.
  - Offline Conversion:** You can losslessly strip the Dolby Vision enhancement layer and metadata offline to leave the standard HDR10 base layer. When you select a Profile 7 file, the server will print a ready-to-run `ffmpeg` command in the console:

```
ffmpeg -i "Movie.mkv" -map 0 -c copy -bsf:v "filter_units=remove_types=62|63" "Movie_no_
```

Run this command on your PC. The output `Movie_no_dv.mkv` will be created losslessly without re-encoding, preserving all audio and subtitle tracks, and can be played perfectly on the client in HDR10 mode.

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## 6. Troubleshooting

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If you encounter issues during playback, check these logs located next to the server executable on your PC:

- client-logs.txt** : Contains playback logs sent dynamically from the Fire TV client. Use this to verify codec selection and audio track initialization.
- server.log** : The server's own logs — requests served, the full metadata dump per file, and warnings (e.g. missing `ffmpeg` / `ffprobe` ).

- `ffmpeg-logs.txt` : Contains detailed output from background chapter/scrubbing thumbnail generation.